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Write a Python Program to Implement Expectation & Maximization Algorithm

Code :

```
import numpy as np

X = np.array([1, 2, 3, 8, 9, 10])
m = np.array([2.0, 8.0])

for i in range(2):

    # E-Step

    p1 = np.exp(-(X - m[0]) ** 2)
    p2 = np.exp(-(X - m[1]) ** 2)
    r = p1 / (p1 + p2)

    # M-Step

    m[0] = np.sum(r * X) / np.sum(r)
    m[1] = np.sum((1 - r) * X) / np.sum(1 - r)

    print("Iteration:", i + 1)

    print("Means:", m)

print("Final Means:", m)
```

Output :

```
Iteration: 1
Means: [2. 9.]
Iteration: 2
Means: [2. 9.]
Final Means: [2. 9.]

=== Code Execution Successful ===
```